

Experiment 8: Effect of PCA on Regression and Classification Models

Objective

To study the impact of **Principal Component Analysis (PCA)** on the performance of machine learning models.

Students must:

- a) Train models in the original feature space (Without PCA)
- b) Train models in reduced feature space (With PCA)
- c) Compare performance using 5-fold cross-validation

Dataset

Dataset:

PCA_Full_Lab_Dataset_1000_Samples.csv

Features: Academic and performance-related attributes .

Regression Target: Final_Score_Regression

Classification Target: Performance_Level_Classification

Students must report:

- Number of samples and features
- Target distribution
- Preprocessing steps

Preprocessing Steps

1. Handle missing values (if any)
2. Encode categorical target (for classification)
3. Standardize features (Very Important before PCA)
4. Split data using 5-fold cross-validation

Graphs and Plots

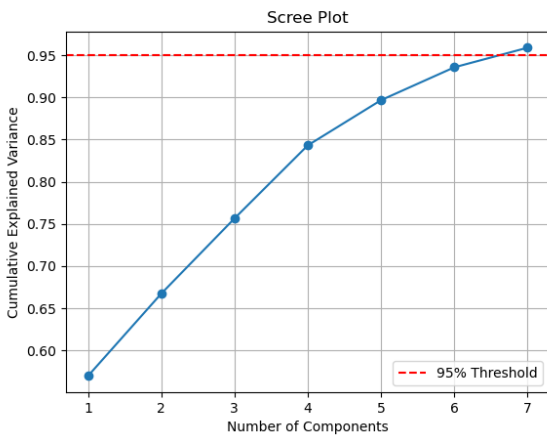


Figure 1: Scree plot

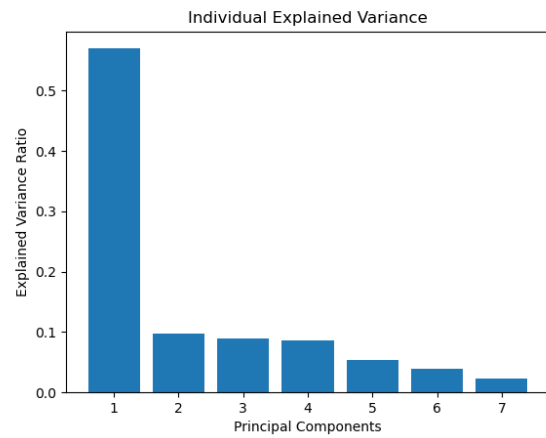


Figure 2: Individual Explained Variance

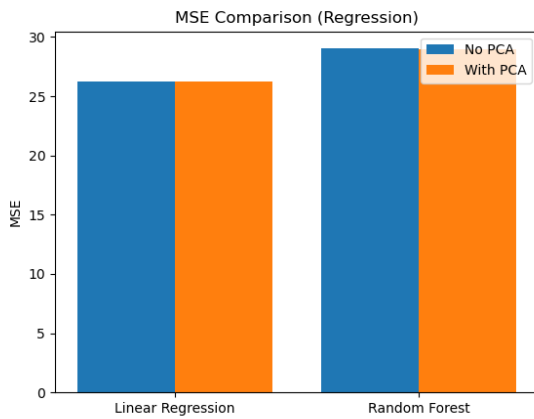


Figure 3: MSE Comparison (Regression)

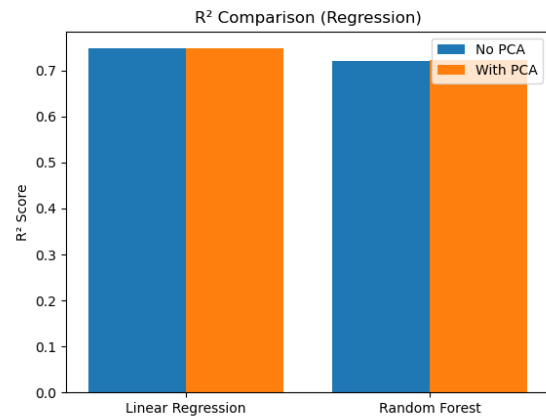


Figure 4: R² Comparison (Regression)

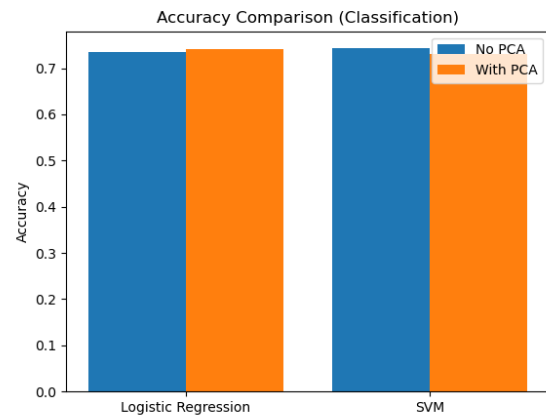
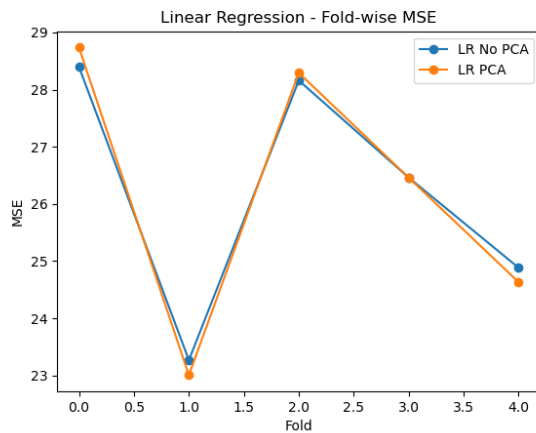


Figure 5: Linear Regression - Fold-wise MSE Figure 6: Accuracy Comparison (Classification)

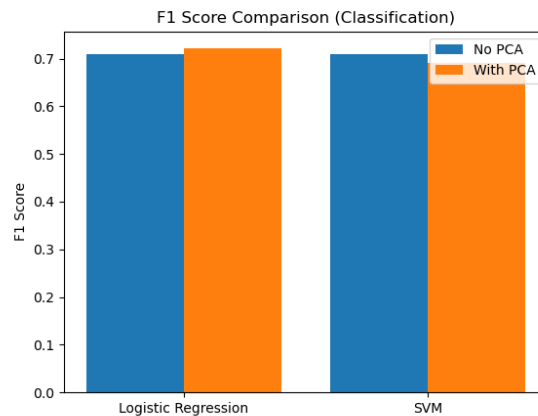


Figure 7: F1 Score Comparison (Classification)

PCA Implementation

1. Apply PCA after standardization
2. Choose:
 - Either fixed number of components
 - Or retain 95% explained variance
3. Plot Scree Plot
4. Record explained variance

Regression Models

1. Linear Regression

Evaluation Metric:

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7	95.90	Retains 95% variance with reduced dimensionality and noise.

- Mean Squared Error (MSE)
- R^2 Score

2. Random Forest Regressor

Hyperparameters to Tune:

- Number of Trees
- Maximum Depth

Regression Results Table

Table 2: 5-Fold CV Results – Regression

Model	Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Linear Regression	MSE	26.2338	26.2251	Slight improvement with PCA
Linear Regression	R^2	0.7479	0.7481	Slight improvement with PCA
Random Forest	MSE	29.0245	28.9563	Negligible improvement
Random Forest	R^2	0.7215	0.7222	Almost unchanged

Classification Models

1. Logistic Regression

Hyperparameter:

- Regularization parameter C

Metrics:

- Accuracy
- F1-score

2. Support Vector Machine (SVM)

Hyperparameters:

- Kernel (Linear / RBF)
- C
- Gamma

Classification Results Table

Table 3: 5-Fold CV Results – Classification

Model	Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Logistic Regression	Accuracy	0.7340	0.7420	Improved with PCA
Logistic Regression	F1-score	0.7093	0.7211	Improved with PCA
SVM	Accuracy	0.7430	0.7300	Decreased with PCA
SVM	F1-score	0.7084	0.6915	Decreased with PCA

Learning Outcome

- PCA reduces multicollinearity
- PCA helps linear models more
- Tree-based models may not benefit much
- PCA reduces noise
- Dimensionality reduction can improve generalization

Final Conclusion

- **When PCA is useful:** PCA is beneficial when dealing with high-dimensional data with correlated features, as it reduces multicollinearity and noise, improving linear model performance.
- **When PCA is not necessary:** PCA is not essential for models like Random Forest that can inherently handle feature redundancy and interactions.
- **Trade-off between interpretability and performance:** While PCA can improve generalization, it transforms features into principal components, reducing interpretability of the original data.